

ValveTrack — Process Flow Update

Response to Requested Changes & Delivery Plan

This document confirms our understanding of the six changes you requested for the Job Card system, explains exactly what we will deliver for each, answers your question about how Accounts works, and sets out a delivery approach designed so that **nothing in the current live system breaks** while the new capabilities are added.

At a glance

- **Due Date & Aging** — with Green / Yellow / Red highlighting across the job list and dashboard
- **Enforced process flow** — six ordered operations that cannot be skipped without authorisation
- **Machine assignment** — welding vs. machining machines assigned per operation
- **Expanded process records** — machine, operator, times, planned / completed / rejected quantity per step
- **Job Card format** — matched to your traveller sheet
- **Dimension Report** — confirmed against your field list
- **Accounts** — explained in full (already built into the system)

Your Requests — and What We Will Deliver

Each item below maps a request from your document to the concrete change we will make.

1. Due Date & Aging

Every job will carry a **Due Date**. The system will automatically show the Due Date, Overdue Days (once the date passes), and Job Aging (days since the Job Card was created). Jobs will be colour-coded in the list and on the dashboard:

- **Green** — On Schedule
- **Yellow** — Due today or nearing the due date
- **Red** — Overdue

2. Enforced Process Flow

The manufacturing sequence will be enforced in the exact order you specified:

Job Card → WPS → Pre-Machining → Welding → Final Machining → Milling → Slitting → Deburring → Quality Inspection → Dispatch

Each job automatically receives only the operations it needs (a machining-only job will not carry an empty Welding step). An operation **cannot be started until the previous one is completed**. If a step genuinely must be done out of order, only an Administrator can authorise it, and that authorisation is recorded. This rule is enforced both on screen and in the database, so it cannot be bypassed.

3. Job Card Format

The generated Job Card PDF will be aligned to your traveller sheet, and will now also include the operations table (each step with its machine, operator, quantities and status). All existing sections are retained.

4. Machine Assignment

A machine list will be maintained in the system, split into **Welding Machines** and **Machining Machines**. Welding machines can be assigned only to the Welding operation; machining machines to Pre-Machining, Final Machining, Milling, Slitting and Deburring.

5. Process Execution

Each of the six operations will have its own execution record capturing:

Field	Field	Field
Machine	Operator	Start Time
End Time	Planned Quantity	Completed Quantity
Rejected Quantity	Status	Remarks

Welding additionally keeps its full WPS-versus-Actual welding parameters, exactly as today.

6. Dimension Report

The Dimension Report already captures every field you listed — Inspection Date, Characteristic, Nominal Dimension, Tolerance, Actual Measurement, Pass/Fail Status, Inspector Name and Inspection Remarks. We will align the on-screen labels to match your terminology exactly.

Your Question: “How do Accounts work?”

The **Accounts** stage is already a complete receivables module. When a job is dispatched and moves to **Accounts Processing**, the Accounts team records the commercial and payment details against that job:

What Accounts records	Purpose
PO Number & PO Value	Links the job to the customer purchase order and its value
Invoice Number, Date & Value	The invoice raised for the dispatched job
GRN Status & Date	Goods Receipt Note confirmation from the customer
Payment Status, Date & Amount	Pending / Partial / Received, with the amount and date
Payment Due Date	When payment is expected
Tally Reference	Cross-reference to your Tally accounting entry
Notes	Any commercial remarks

A job **cannot be Closed** until it has been dispatched and (for non-admin users) payment status shows **Received**. An Administrator can override the payment condition in exceptional cases. In short: Accounts turns a completed, dispatched job into a tracked, billed, and paid job before it is closed.

How We Will Deliver This Safely

This is an enterprise-grade update, not a quick patch. Our approach protects the live system throughout:

- **Additive only.** New fields and tables are added; no existing data or Job Card is altered or invalidated. Every current job card keeps working and printing.
- **Tested on a separate copy first.** All database changes are built and tested on an isolated preview copy of the system, then merged — production is never edited by hand.
- **Verified at every step.** After each change the system is rebuilt, type-checked, and a full end-to-end test job is run to confirm existing behaviour is intact.
- **Reversible.** A verified baseline and rollback point is established before any change is made.

Delivery Phases

Phase	Scope	Effort
0	Safety baseline & rollback point	Small
1	Machine master & assignment	Small
2	Six-operation routing per job	Large
3	No-skip sequence enforcement	Medium
4	Due Date & Aging with colour coding	Medium
5	Job Card PDF alignment	Medium
6	Dimension Report labels + Accounts walkthrough	Small
7	Full quality assurance & regression	Medium

Phases build on one another; Due Date & Aging can be delivered in parallel.

What We Need From You

- Confirmation to proceed with this plan.

- Your machine list (welding and machining machines) to pre-load the master.
- Your Job Card sample sheet, if it differs from the traveller we already have, for exact PDF matching.